

The price-tag decision: logistic regression and menu costs at a large retail store

Javier Brito Martillo and Luis Paredes | Peer programming exercise, Problem 1

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Data: retail_relabel.csv (1,500 SKUs, no missing values). Code: retail_relabel_analysis.py, runs end to end, random_state = 7.

Part 1. Feature engineering and exploratory analysis

We averaged the three competitor price columns into `CompAvg`, then built the signed gap $\text{GapPct} = 100 * (\text{StorePrice} - \text{CompAvg}) / \text{CompAvg}$, its absolute value `AbsGapPct`, $\text{LogUnits} = \ln(\text{WeeklyUnitsSold})$, and $\text{MarginPct} = 100 * (\text{StorePrice} - \text{UnitCost}) / \text{StorePrice}$. Velocity enters in logs because the amortization benefit of a fixed menu cost rises with volume at a decreasing rate. Doubling weekly units matters far more for an item selling 10 a week than for one selling 100.

Feature	Mean	Std dev	Min	Median	Max
GapPct	3.62	12.73	-28.27	3.71	42.17
AbsGapPct	11.00	7.36	0.02	10.10	42.17
LogUnits	3.94	0.50	1.79	3.99	5.16
MarginPct	32.68	7.74	16.73	33.74	44.44

The overall relabel base rate is **40.0 percent** (600 of 1,500 tags changed). Category variation is mild: Electronics 44.0 percent, Grocery 41.2 percent, Apparel 39.0 percent, Personal Care 38.3 percent, Household 37.8 percent. Nothing here suggests category is doing much independent work once the economics are in the model.

The ESL split is where the story is

Shelf label type	SKUs	Mean relabel cost	Relabel rate
Paper tag (ESL = 0)	1,045	\$1.45	33.9%
Electronic label (ESL = 1)	455	\$0.06	54.1%

Items on electronic labels are corrected at roughly **1.6 times** the rate of paper items, and the only thing separating the two groups is the price of pressing the button. The menu cost is not a rounding error in this store, it is a filter that decides which mispriced items ever get fixed. A paper tag has to be wrong by enough to justify \$1.45 of labor and materials before anyone walks over with a label gun. An electronic tag only has to be wrong.

Mispricing itself lines up with the outcome the way theory says it should. SKUs that were relabeled sat an average of **15.1 percent** away from the competitor average, against **8.3 percent** for those left alone.

Part 2. Model 1, the naive signed-gap model

`Relabel ~ GapPct + UnitCost + RelabelCost`

Term	Coefficient	Std error	z	p
Intercept	0.0589	0.098	0.60	0.547
GapPct	0.0379	0.005	8.38	< 0.001
UnitCost	0.0005	0.003	0.15	0.877
RelabelCost	-0.6387	0.071	-9.03	< 0.001

Pseudo R-squared = 0.0773 | AIC = 1,871.02 | BIC = 1,892.27 | n = 1,500

Why the signed gap is the wrong functional form

Ask the owner what she does about an item priced 10 percent below the three rivals. She raises the price and captures the margin she was giving away. Now ask what she does about an item priced 10 percent above them. She cuts the price before the shopper walks out with a competitor's app open. Both items get a new tag. The action is the same, only the direction of the correction differs.

The signed gap cannot express that. A logit coefficient on `GapPct` is a single monotonic slope: it says the probability of a tag change either always rises or always falls as the gap moves from negative to positive. The truth is U-shaped, and the data show it plainly.

Signed gap bucket	-20% and below	-20 to -10	-10 to -5	-5 to 0	0 to 5	5 to 10	10 to 20	20% and above
Relabel rate	70.0%	48.5%	31.4%	12.2%	14.6%	26.6%	54.9%	79.9%
SKUs	20	237	172	188	192	188	344	159

Relabeling is common at both ends and rare in the middle. Forcing one slope through that shape averages the two arms of the U against each other, and the estimate that survives is a diluted version of the effect on the overpriced side, where more of the mass happens to sit. The single slope is still significant, which is the trap. Significance does not mean the form is right.

Holding the other two predictors fixed and swapping only `GapPct` for `AbsGapPct`, with no new variables added, pseudo R-squared rises from **0.0773 to 0.2206** and AIC falls from **1,871 to 1,582**. That is the cost of the functional form alone, isolated from every other modeling choice.

Part 3. Model 2, the economically motivated model

Relabel ~ AbsGapPct + CostChangePct + LogUnits + RelabelCost + DaysSinceLastChange + UnitCost

Term	Coefficient	Odds ratio	95% CI	p
Intercept	-4.2772	0.014	[0.005, 0.042]	< 0.001
AbsGapPct	0.1778	1.195	[1.169, 1.221]	< 0.001
CostChangePct	0.1276	1.136	[1.099, 1.174]	< 0.001
LogUnits	0.4324	1.541	[1.192, 1.992]	0.001
RelabelCost	-0.8589	0.424	[0.359, 0.501]	< 0.001
DaysSinceLastChange	0.0040	1.004	[1.003, 1.005]	< 0.001
UnitCost	0.0004	1.000	[0.993, 1.008]	0.919

Pseudo R-squared = 0.2750 | AIC = 1,477.82 | BIC = 1,515.01 | n = 1,500

What the three key coefficients mean for the owner

- **AbsGapPct, odds ratio 1.195.** Every extra percentage point of distance from the competitor average, in either direction, raises the odds of a tag change by about 20 percent. Ten points of misalignment multiplies the odds by roughly 5.2. Being wrong is what triggers the correction, not being expensive.
- **RelabelCost, odds ratio 0.424.** Each additional dollar of menu cost cuts the odds of a correction by about 58 percent. This is the single strongest deterrent in the model, and it is entirely under management control.
- **LogUnits, odds ratio 1.541.** A one-unit rise in log units, about a 2.7 times increase in velocity, raises the odds of correction by roughly 54 percent. Fast movers earn back the label cost sooner, so they get fixed first.

Why CostChangePct matters and the level of UnitCost does not

CostChangePct carries an odds ratio of 1.136 at $p < 0.001$, while UnitCost sits at 1.000 with $p = 0.919$, a confidence interval straddling 1, and no predictive content at all. This is exactly the pattern deductive economics predicts.

A tag reflects the price that was optimal when it was printed. What makes it stale is a *change* in conditions since then. A supplier cost increase of 8 percent since the last labeling squeezes margin now, so the tag no longer matches the item's economics and repricing restores the target markup. The level of unit cost carries no such information. A \$2 can of beans and a \$200 blender can both be perfectly priced, or both badly priced. Knowing that an item is expensive tells you nothing about whether its tag is wrong, because the price was already set with that cost in view. Changes create the misalignment. Levels do not. The insignificant coefficient on UnitCost is a confirmed prediction, not a failure to find something.

Model 1 against Model 2

Metric	Model 1 (signed gap)	Model 1b (gap swapped only)	Model 2 (full)
Pseudo R-squared	0.0773	0.2206	0.2750
AIC	1,871.02	1,581.64	1,477.82
BIC	1,892.27	1,602.89	1,515.01

Model 2 wins on all three, and it wins on BIC as well, which penalizes the three extra parameters harder than AIC does. The improvement is not bought with variable count. Roughly three quarters of the gain in pseudo R-squared comes from correcting the functional form of the gap, and the rest from adding the cost-pressure, velocity, and staleness terms that the theory said belonged there. **We would deploy Model 2.**

Part 4. Out-of-sample evaluation

A 70/30 stratified split with random_state = 7 gives 1,050 training and 450 test SKUs, each holding the 40.0 percent base rate. Refitting Model 2 on the training rows only (Model 3) and scoring the held-out 450 at a 0.50 threshold:

Test confusion matrix	Predicted 0	Predicted 1	Row total
Actual 0 (no relabel)	233	37	270
Actual 1 (relabel)	64	116	180

Accuracy = 0.776 | AUC = 0.853 | Precision = 0.758 | Recall = 0.644 | Specificity = 0.863 | Training pseudo R-squared = 0.2552

Accuracy of 77.6 percent beats the 60 percent you would get by predicting "no relabel" for everything, and an AUC of 0.853 means that if you pick one relabeled SKU and one untouched SKU at random, the model ranks them correctly about 85 percent of the time. The error profile is worth naming. The model misses 64 tags that were actually changed and flags 37 that were not, so at a 0.50 cutoff it is conservative: it under-calls corrections. If the owner cares more about catching mispriced items than about avoiding a few unnecessary label prints, and the payback numbers in Part 5 say she should, then the cutoff belongs below 0.50, not at it. The threshold is a business choice, not a statistical constant.

Part 5. Economic decision analysis

For every test SKU we computed $\text{ExpectedWeeklyLoss} = (\text{AbsGapPct} / 100) * \text{StorePrice} * \text{WeeklyUnitsSold}$ and $\text{NetBenefit} = \text{ExpectedWeeklyLoss} - \text{RelabelCost}$.

444 of 450 test SKUs, or 98.7 percent, recoup the relabel cost within a single week of corrected pricing. The median expected weekly loss is \$43.18 against a menu cost of at most a few dollars. Correcting every SKU with a positive net benefit is worth about \$60,735 per week on the test set alone.

Segment	Test SKUs	Mean AbsGapPct	Mean net benefit / week	Payback within a week
Electronics	60	11.4%	\$533.90	100.0%
Apparel	85	11.3%	\$159.35	100.0%
Household	80	9.4%	\$78.35	98.8%
Personal Care	75	9.8%	\$47.16	97.3%
Grocery	150	10.4%	\$35.65	98.0%

Misalignment is spread fairly evenly across categories, so the ranking is driven by price times velocity, not by which aisle is worst priced. Electronics carries fifteen times the dollar value per SKU of Grocery at almost the same percentage gap.

A useful caution before anyone treats these dollars as profit. $\text{ExpectedWeeklyLoss}$ values the entire price gap as recoverable revenue, which it is not. Part of any gap is deliberate positioning, shoppers do not switch stores over every cent, and an underpriced item that gets a higher tag may sell fewer units. Read the number as the value at stake, an upper bound on what correct pricing is worth, and apply a haircut before it hits a budget. Even so, the ratio of value at stake to menu cost is so lopsided, roughly 30 to 1 at the median, that the ranking survives any reasonable discount.

Memo to the store owner

To: Owner, SunCoast Retail Mart **From:** Javier Brito Martillo and Luis Paredes **Re:** Repricing policy from this cycle's data

Which items to relabel first. Build the queue on three signals, in this order. First, distance from the competitor average in either direction, since every extra point of gap raises the odds a tag needs changing by about 20 percent, and being priced too low costs you as much margin as being priced too high costs you traffic. Second, weekly velocity, because a fixed label cost is recovered faster on items that move. Third, cost pressure: any SKU whose supplier cost moved since its last tag is now carrying a margin you did not choose. In practice that means an operational queue of **Electronics and Apparel SKUs with a gap above 10 percent and healthy weekly volume**, where the average item returns \$534 and \$159 per week respectively against a menu cost under \$2. Applying the model to the 450 held-out SKUs produced 153 items that are both likely to need a change and profitable to change, worth about **\$37,861 per week**. Slow-moving, low-price, well-aligned items can wait for the next cycle without much cost. The level of an item's unit cost should not be on the list. It carries no information about whether the tag is wrong.

The case for converting more shelves to electronic labels. Today 1,045 of your 1,500 SKUs sit on paper tags at an average change cost of \$1.45. The 455 on electronic labels cost \$0.06 to change, and they are relabeled at 54.1 percent per cycle against 33.9 percent for paper. The model says this gap is caused by the cost itself, not by which items happen to have electronic tags: holding every other characteristic of the paper-tag SKUs fixed and lowering only their menu cost to the electronic level, predicted relabel probability rises from 0.34 to 0.54. That is the observed electronic rate, reproduced from the cost change alone. In practice that means about **208 additional mispriced items corrected every cycle**. Those items currently carry \$27,064 per week of gross misalignment. Even assuming you capture only a tenth of that once behavior, elasticity, and deliberate positioning are accounted for, conversion is worth roughly **\$140,000 a year, or about \$135 per converted shelf per year**. That is the number to hold up against a hardware quote. Right now, \$65,428 of weekly gross misalignment is sitting on 691 paper-tag items that were left alone, and the cheapest way to fix that is to make fixing it cheap.

What this analysis cannot tell you. These are observational data from one pricing cycle. Nobody randomly assigned electronic labels to shelves, so the effects above are associations, not proven causes. If electronic labels went onto your busiest or most competitive aisles first, part of what we are reading as a menu-cost effect is really an aisle effect, and the model has no way to separate the two. The conclusion is directionally strong and consistent with basic price theory, but before committing capital across the whole store we would convert one or two aisles chosen at random, hold the rest as a control, and measure the change in relabel rates and margin over a few cycles. That turns a defensible association into a number you can bank on. We would also note that the model was fit on past behavior, so it predicts what your team *did* decide, not necessarily what they *should* have decided.

Discussion questions from the code walkthrough

1. Model 1 against Model 2, and what the signed gap hides

Signing a variable that should be measured symmetrically is the core mistake this assignment is built around. The owner's decision rule is symmetric: fix the tag when it is far from the market, whichever side it is on. `GapPct` encodes direction, which the decision does not depend on, and discards distance, which is the only thing it does depend on. A single logit slope then has to approximate a U-shaped relationship with a straight line in the log-odds, so the two arms partially cancel and what survives is a weakened version of the real effect. The evidence is in the fit. Model 1 reaches pseudo R-squared 0.0773 with AIC 1,871.02. Model 2 reaches 0.2750 with AIC 1,477.82. To rule out the objection that Model 2 simply has more variables, we refit Model 1 with the identical three predictors and only the gap's form changed: pseudo R-squared 0.2206, AIC 1,581.64. The functional form by itself accounts for about three quarters of the improvement. The dilution is also visible before any model is fit, in the bucket table on page 1, where the relabel rate runs 70 percent, falls to 12 percent near zero gap, then climbs back to 80 percent. Model 1 was statistically significant the whole time. Significance told us the coefficient was not zero. It did not tell us the variable was the right one.

2. MarginPct is computed and never used. Should it have been?

We tested it rather than guessing. Adding `MarginPct` to Model 2 gives a coefficient of -0.0011 with $p = 0.892$, leaves pseudo R-squared unchanged at 0.2750, and raises AIC from 1,477.82 to 1,479.80. A likelihood-ratio test returns $p = 0.892$. It contributes nothing, and there are two reasons why.

First, it is nearly redundant. `MarginPct` correlates 0.897 with `GapPct` in this data, which makes sense: given roughly stable costs, an item priced above its competitors is mechanically an item carrying a fat margin. The information is already in the model in a better form, because `AbsGapPct` measures it symmetrically and `MarginPct` does not. Second, and more important, margin is downstream of the decision, not upstream of it. Current margin is a consequence of the tag that is already on the shelf. What makes a tag wrong is a change in the conditions that set it, which is why `CostChangePct` is significant and both `UnitCost` and `MarginPct` are not.

That said, we would not throw it away. `MarginPct` earns its place in the economic analysis rather than the model. A symmetric version, distance from the target markup for the category, would be a legitimate predictor and would capture margin drift that competitor prices miss. And in a version of this problem where the objective is profit rather than the historical relabel decision, margin is the whole point: it tells you how much money is riding on each correction, which is precisely what we used the gap for in Part 5. Computing it and then leaving it out of every model is worth flagging as a loose end in the walkthrough, not as a silent choice.

3. Section 6 against Sections 4 and 5: fit against generalization

These sections answer two different questions and it is easy to blur them. Pseudo R-squared 0.2750 and AIC 1,477.82 in Sections 4 and 5 describe how well Model 2 fits the 1,500 rows it was estimated on. They are in-sample quantities. AIC even admits this in its construction, penalizing parameters precisely because more parameters always fit the training data better. Section 6 asks a different question: does the model work on SKUs it has never seen? Refit on 1,050 training rows and scored on 450 held-out rows, it delivers 77.6 percent accuracy and an AUC of 0.853.

The two sets of numbers are not comparable, and no amount of pseudo R-squared would substitute for the second. Pseudo R-squared cannot tell you whether the model overfit, because it is computed on the same data that produced the coefficients. Training pseudo R-squared drops from 0.2750 on the full sample to 0.2552 on the training subset, and the model still generalizes cleanly, which is the reassuring outcome. Had the test AUC collapsed while the in-sample fit stayed high, we would have learned something the in-sample statistics were structurally incapable of showing. The practical difference is which promise you can make. In-sample fit lets you say the model describes what happened last cycle. Out-of-sample evaluation lets you say it will rank next cycle's SKUs. Only the second one supports an operational decision, and notice which one the AUC of 0.853 is.

4. Section 7: significance is not a business case

By the end of Section 6 we have a model that is statistically significant on five of six predictors, fits reasonably, and generalizes. None of that tells the owner what to do on Monday morning. A coefficient is a statement about odds. A store runs on dollars.

Section 7 is the translation layer, and it changes the recommendation in ways the model alone never could. The model, at a 0.50 threshold, predicts a relabel for about 34 percent of test SKUs. The economics say 98.7 percent of them would recoup the relabel cost within a single week. Those two statements point in different directions, and the second one is the one that pays. It says the store's revealed threshold for repricing has been too conservative, and that the 0.50 cutoff we inherited from convention is not the cutoff a profit-maximizing owner would pick. It also reranks the work: `AbsGapPct` barely varies across categories, so nothing in the model singles out Electronics, but Electronics carries \$534 of weekly value per SKU against Grocery's \$36, because net benefit is a function of price and volume as well as misalignment. And it is what converts the ESL coefficient, an odds ratio of 0.424 per dollar, into an annual figure a hardware quote can be measured against. Without Section 7 the analysis ends with a p-value. With it, the analysis ends with a queue, a budget, and a threshold the owner can argue with.